doc-agent: what was implemented

Complete list of the functions, classes and constants implemented (or added) when the skeleton was completed. Signatures are extracted directly from the source. Files marked **NEW** did not exist before; **IMPL** were stubs that are now implemented; **FIXED** are provided files that were left unchanged (listed in the guide PDF).

Summary

51 source modules implemented, plus scripts, the verifier, config and test files. Run `python -m pytest` to verify (40 tests pass).

Source modules (src/doc_agent)

src/doc_agent/agent/agent.py *[IMPL]*

Stage 6 - FIXED loop - perceive -> decide -> act -> observe, with cross-cutting seams.

Kind	Name / signature	What it does
class	`Agent`	FIXED loop. Implement decide() (the policy) and synthesize() only.
method	`Agent.__init__(self, cfg: dict, retriever) -> None`	
method	`Agent.run(self, query_text: str) -> Answer`	
method	`Agent.decide(self, state: dict) -> dict`	Evidence-gated re-search — the MANDATORY agentic behaviour (A3 gate, fail-closed).
method	`Agent.act(self, action: dict) -> ToolResult`	
method	`Agent.synthesize(self, state: dict) -> Answer`	Grounded, cited answer; abstain if unsupported (no-hallucination).

src/doc_agent/agent/guardrails.py *[IMPL]*

Stage 6 — SECURITY — autonomy, budgets, prompt-injection defense

Kind	Name / signature	What it does
constant	`INJECTION_PATTERNS`	
constant	`_INJ`	
constant	`SIDE_EFFECT_TOOLS`	
function	`looks_like_injection(text: str) -> bool`	
function	`sanitize(text: str) -> str`	Neutralise injection phrases in untrusted document text. Length-preserving (masks with '~')
function	`_strings(obj)`	
class	`Guardrails`	Enforce autonomy level, step/cost budget, and instruction/content isolation.
method	`Guardrails.__init__(self, cfg: dict) -> None`	
method	`Guardrails.reset(self) -> None`	
method	`Guardrails.check(self, action: dict) -> None`	Raise PermissionError if: tool unknown, step budget or $ budget exceeded, a side-effect
function	`register(hooks, cfg: dict) -> None`	Wire guardrails into every tool call (and reset budgets at the start of each query).

src/doc_agent/agent/hitl.py *[IMPL]*

HITL — human-in-the-loop review queue

Kind	Name / signature	What it does
function	`escalate(reason: str, context: dict) -> ToolResult`	Queue for human review. Returns status 'pending': the caller must NOT act until
function	`review_queue()`	Return pending items for the reviewer UI.

src/doc_agent/agent/hitl_store.py *[IMPL]*

HITL — persistent review queue (survives restarts)

Kind	Name / signature	What it does
constant	`DB_PATH`	
function	`_conn() -> sqlite3.Connection`	
function	`enqueue(item: dict) -> str`	Persist a pending review item; return its id.
function	`pending() -> list[dict]`	
function	`resolve(item_id: str, decision: str) -> None`	decision: 'approved' | 'rejected'.
function	`decision(item_id: str) -> str | None`	None while still pending, else 'approved'/'rejected' (used to block actions until reviewed).

src/doc_agent/agent/memory.py *[IMPL]*

Stage 6 — working/episodic memory

Kind	Name / signature	What it does
function	`_text(item) -> str`	Searchable text of a memory item (ToolResult payloads, dicts, plain strings).
class	`Memory`	
method	`Memory.__init__(self) -> None`	
method	`Memory.add(self, item) -> None`	
method	`Memory.recall(self, query: str, top: int=3) -> list`	Items ranked by word overlap with the query (working-memory lookup, no embeddings).

src/doc_agent/agent/tools.py *[IMPL]*

Stage 6 — FIXED tool interface — the agent's tools

Kind	Name / signature	What it does
function	`bind(retriever, cfg: dict) -> None`	
function	`_retriever()`	
function	`_chunk(chunk_id: str) -> Chunk | None`	
class	`Tool(ABC)`	
class	`Retrieve(Tool)`	
class	`Rerank(Tool)`	
class	`ReadPage(Tool)`	
class	`EnhancePage(Tool)`	
constant	`_NUM`	
constant	`_PATTERNS`	
class	`Extract(Tool)`	
class	`Aggregate(Tool)`	
class	`Cite(Tool)`	
constant	`_OPS`	
function	`_eval(node)`	
class	`Calculator(Tool)`	
class	`EscalateToHuman(Tool)`	
constant	`REGISTRY`	

src/doc_agent/data/validate.py *[IMPL]*

Data — data schema/quality validation at ingest

Kind	Name / signature	What it does
constant	`VALID_EXTS`	

Kind	Name / signature	What it does
function	`validate(pages: list[Page]) -> None`	Structural checks (raise AssertionError): non-empty, unique ids, readable image files of a
function	`split_by_doc(pages: list[Page], seed: int=42, frac=(0.8, 0.1, 0.1)) -> dict[str, list[Page]]`	Split by DOCUMENT (never by page) so no document leaks across train/val/test.
function	`assert_no_leakage(splits: dict[str, list[Page]]) -> None`	
function	`check_floors(n_pages: int, n_words: int, task: dict) -> dict`	Compare against corpus floors in configs/task.yaml. Returns a report (does not raise).
function	`file_hash(path: str | Path) -> str`	

src/doc_agent/data/versioning.py *[IMPL]*

Data — corpus versioning (which corpus version -> which result)

Kind	Name / signature	What it does
function	`snapshot(corpus_dir: str) -> str`	Content-hash every file under corpus_dir (path + bytes) -> 12-char version id, and append

src/doc_agent/eval/ablation.py *[IMPL]*

Stage 9 — ablation harness

Kind	Name / signature	What it does
constant	TASKS	
function	`load_tasks(path: str=TASKS) -> list[dict]`	Eval items; placeholder rows with an empty question are skipped.
function	`evaluate(cfg: dict, tasks: list[dict]) -> dict`	Run every task through the full agent and aggregate the headline metrics.
function	`run(cfg: dict) -> dict`	Toggle each stage off; report metric deltas vs the full system. Variants are query-time

src/doc_agent/eval/calibration.py *[IMPL]*

Stage 9 — confidence calibration (calibrated-confidence NFR)

Kind	Name / signature	What it does
class	`TemperatureScaler`	softmax(logits / T). T > 1 softens over-confident models, T < 1 sharpens.
method	`TemperatureScaler.__init__(self, temperature: float) -> None`	
function	`temperature_scale(logits, labels)`	Fit temperature on a validation set by minimising NLL (coarse log-grid, then refine).
function	`ece(confidences, correct) -> float`	Expected calibration error with 10 equal-width bins: sum_b (n_b/n)*|acc_b - conf_b|.

src/doc_agent/eval/fairness.py *[IMPL]*

Stage 9 — subgroup audit

Kind	Name / signature	What it does
function	`audit(cfg: dict) -> dict`	Does the system serve all scan-quality subgroups equally? Pages are bucketed (good / medium /

src/doc_agent/eval/interpret.py *[IMPL]*

Stage 9 — EXPLAINABLE — why-retrieved / where-looked

Kind	Name / signature	What it does
function	`explain(answer: Answer, cfg: dict) -> dict`	Per citation: WHY it was retrieved (cosine similarity of the answer to the chunk plus the
function	`_regions(page_ids: list[str], cfg: dict) -> dict[str, list]`	

src/doc_agent/eval/judge.py *[IMPL]*

Stage 9 — LLM-as-judge for non-verifiable inference

Kind	Name / signature	What it does
function	`judge(query: Query, answer: Answer) -> float`	Score an open-ended answer in [0, 1]. With an LLM key: the JUDGE prompt, parsed as a float.

src/doc_agent/eval/metrics.py *[IMPL]*

Stage 9 — metrics

Kind	Name / signature	What it does
function	`ocr_f1(pred: str, gold: str) -> float`	Token-level F1 between OCR output and ground-truth transcription (bag of words).
function	`recall_at_k(retrieved: list, gold: list, k: int) -> float`	Fraction of gold items found in the top-k retrieved (1.0 if there is no gold).
function	`groundedness(answer: Answer) -> float`	1.0 if the answer is consistent: grounded AND cited, or an honest abstention (not grounded,
function	`citation_accuracy(answer: Answer) -> float`	Share of citations that are well-formed and, when the index is available, point at a real
function	`ece(confidences, correct) -> float`	
function	`subgroup_gap(scores_by_group: dict) -> float`	Max minus min of the per-group mean scores (0 = perfectly even).

src/doc_agent/eval/robustness.py *[IMPL]*

Stage 9 — OOD / scan-quality stress

Kind	Name / signature	What it does
constant	`CORRUPTIONS`	
function	`stress(cfg: dict, max_pages: int=5) -> dict`	Corrupt sample pages at severity 1..3 (noise, blur, low contrast, rotation) and measure how

src/doc_agent/governance/pii.py *[IMPL]*

Governance — PII detection + redaction (mandatory)

Kind	Name / signature	What it does
constant	`_PATTERNS`	
constant	`_ORDER`	
function	`detect(text: str) -> list[tuple[int, int, str]]`	Return non-overlapping (start,end,type) PII spans, sorted by position.
function	`redact(text: str) -> str`	Replace each PII span with [TYPE].
function	`_scrub_obj(obj)`	Recursively redact strings in dicts/lists; returns the cleaned object.
function	`register(hooks) -> None`	Wire PII redaction into the pipeline. The handler works IN PLACE on whatever the seam passes:

src/doc_agent/index/chunk.py *[IMPL]*

Stage 4 — chunk text

Kind	Name / signature	What it does
function	`split(chunks: list[Chunk], cfg: dict) -> list[Chunk]`	Re-chunk to cfg['index'] size/overlap: concatenate the (line-level) OCR chunks of each

src/doc_agent/index/embed.py *[IMPL]*

Stage 4 — embed chunks

Kind	Name / signature	What it does
function	`get_model(name: str)`	Cached SentenceTransformer (loading takes seconds, so do it once per process).

Kind	Name / signature	What it does
function	`encode_texts(texts: list[str], cfg: dict) -> np.ndarray`	L2-normalised float32 embeddings, so inner product == cosine similarity.
function	`encode(chunks: list[Chunk], cfg: dict)`	Embed with cfg['embed']['model'] -> array of shape (n_chunks, dim).

src/doc_agent/index/store.py *[IMPL]*

Stage 4 — vector store

Kind	Name / signature	What it does
function	`_new_index(kind: str, dim: int)`	cfg['index']['type']: 'faiss:hnsw' (approximate, graph) or 'faiss:flat' (exact).
function	`build(chunks, vectors, cfg: dict) -> None`	Persist a vector index (cfg['index']['type']) + the chunk records next to it.
function	`load(cfg: dict)`	Return (faiss_index, list[Chunk]); row i of the index <-> chunks[i].

src/doc_agent/ingest/enhance.py *[IMPL]*

Stage 1 — ENHANCEMENT (VAE / diffusion) — generative denoise / super-resolution of degraded scans

Kind	Name / signature	What it does
class	`DenoiseNet(nn.Module)`	Small residual conv encoder-decoder ('unet_small'). Trained as a one-step denoiser
method	`DenoiseNet.__init__(self, ch: int=16) -> None`	
method	`DenoiseNet.forward(self, x: torch.Tensor) -> torch.Tensor`	
function	`degrade(clean: np.ndarray, rng: np.random.Generator) -> np.ndarray`	Synthetic scan damage (float [0,1] array): blur + gaussian noise + salt/pepper speckle.
function	`cut_patches(pages: list[Page], patch: int, n_per: int, seed: int) -> list[np.ndarray]`	Random clean grayscale patches (float [0,1]) from the pages.
class	`Enhancer`	Model set by cfg['enhance']. train() fits the denoiser on synthetic degradations of the
method	`Enhancer.__init__(self, cfg: dict) -> None`	
method	`Enhancer.train(self, pages: list[Page]) -> None`	
method	`Enhancer.enhance_image(self, gray: np.ndarray) -> np.ndarray`	
method	`Enhancer.apply(self, pages: list[Page]) -> list[Page]`	
function	`run(pages: list[Page], cfg: dict) -> list[Page]`	

src/doc_agent/ingest/loader.py *[IMPL]*

Stage 1 — load scanned page-images

Kind	Name / signature	What it does
constant	`IMAGE_EXTS`	
constant	`_DEFAULT_PATHS`	
function	`data_paths(cfg: dict) -> dict[str, Path]`	Resolve every filesystem location from cfg['data'] (defaults match configs/config.yaml).
function	`_doc_id(path: Path, raw_dir: Path) -> str`	Sub-folder name if the page sits in one, else the file stem minus a trailing page number.
function	`load_pages(cfg: dict) -> list[Page]`	Read data/raw/ -> list[Page] (sorted, so every run sees the same order).

src/doc_agent/ingest/preprocess.py *[IMPL]*

Stage 1 — deskew / denoise / binarize / augment

Kind	Name / signature	What it does
function	`read_gray(path: str) -> np.ndarray`	
function	`write_image(path, img: np.ndarray) -> None`	
function	`estimate_skew(gray: np.ndarray) -> float`	Skew angle in degrees from the min-area rectangle around ink pixels.
function	`quality_stats(gray: np.ndarray, skew: float) -> dict`	Scan-quality metadata (used by fairness audit, drift monitor, robustness).
function	`run(pages: list[Page], cfg: dict) -> list[Page]`	Deskew -> median denoise -> CLAHE contrast normalisation. Writes to data/interim/preprocessed.

src/doc_agent/llm/client.py *[IMPL]*

LLM — the single LLM call wrapper (all model calls go through here)

Kind	Name / signature	What it does
constant	`_STOP`	
function	`content_words(text: str) -> list[str]`	
class	`LLM`	Model set by cfg['agent'] (llm_model). Key from settings.llm_api_key (.env). Providers:
method	`LLM.__init__(self, cfg: dict) -> None`	
method	`LLM.complete(self, prompt: str, **kw) -> str`	

src/doc_agent/llm/postprocess.py *[IMPL]*

LLM — answer post-process / format / abstention

Kind	Name / signature	What it does
constant	`ABSTAIN_TEXT`	
constant	`MIN_SUPPORT`	
function	`abstain_answer() -> Answer`	
function	`support_score(answer_text: str, evidence_texts: list[str]) -> float`	Lexical support: share of the answer's content words that occur in the cited evidence.
function	`format_answer(raw: str, citations: list) -> Answer`	Attach citations, set grounded/confidence, enforce abstention.
function	`register(hooks) -> None`	Grounding / abstention gate at BEFORE_ANSWER: if there is no retrieval evidence, or the best

src/doc_agent/llm/prompts.py *[IMPL]*

LLM — FIXED prompt template registry (all prompts live here)

Kind	Name / signature	What it does
constant	`ABSTAIN_TOKEN`	
constant	`DECIDE`	
constant	`SYNTHESIZE`	
constant	`JUDGE`	

src/doc_agent/logging_conf.py *[IMPL]*

FIXED — structured logging (auditable NFR). Use get_logger(), never print().

Kind	Name / signature	What it does
function	`get_logger(name: str) -> logging.Logger`	
constant	`TRACE_PATH`	
function	`register(hooks) -> None`	Wire structured tracing at each seam (auditable trail) AND emit traces/run.jsonl.

src/doc_agent/mlops/monitor.py *[IMPL]*

MLOps — drift / latency monitoring

Kind	Name / signature	What it does
function	`_p95_target_ms(task_file: str='configs/task.yaml') -> float | None`	
function	`check_drift(cfg: dict) -> dict`	Input drift + latency SLO.

src/doc_agent/mlops/registry.py *[IMPL]*

MLOps — model registry

Kind	Name / signature	What it does
constant	`REGISTRY_FILE`	
function	`register(component: str, path: str, metrics: dict) -> str`	Version a checkpoint + its metrics: records sha256 of the file and returns 'component:vN'.

src/doc_agent/mlops/tracking.py *[IMPL]*

MLOps — experiment tracking (W&B)

Kind	Name / signature	What it does
constant	`_RUN`	
constant	`_LOCAL`	
function	`init_run(cfg: dict, tags: list[str])`	Start a W&B run and log the config. Without WANDB_API_KEY, W&B is put in 'disabled' mode
function	`log(metrics: dict) -> None`	

src/doc_agent/optional/api_security.py *[IMPL]*

OPTIONAL — API auth + rate limiting

Kind	Name / signature	What it does
function	`_h(token: str) -> str`	
function	`configure(tokens: set[str], rate: float=1.0, burst: float=5.0) -> None`	
function	`require_auth(token: str) -> bool`	Constant-time check of a bearer token against the configured set (deny if none configured).
function	`rate_limit(client: str) -> bool`	Token bucket per client. True = request allowed.

src/doc_agent/optional/cache.py *[IMPL]*

OPTIONAL — embedding/retrieval/answer cache

Kind	Name / signature	What it does
constant	`MAX_ITEMS`	
constant	`DISK`	
function	`get(key: str)`	In-memory LRU first, then the on-disk log. Returns None on a miss.
function	`put(key: str, val) -> None`	

src/doc_agent/optional/config_snapshot.py *[IMPL]*

OPTIONAL — snapshot exact run config for reproducibility+

Kind	Name / signature	What it does
function	`snapshot(cfg: dict, out: str) -> None`	Write cfg + git commit + library versions + corpus version to `out` (JSON).

src/doc_agent/optional/dp.py *[IMPL]*

OPTIONAL — differential privacy (DP-SGD) for shared models

Kind	Name / signature	What it does
class	`DPOptimizer`	Wraps an optimizer: clips the global gradient norm to `clip` and adds Gaussian noise

Kind	Name / signature	What it does
method	`DPOptimizer.__init__(self, opt: torch.optim.Optimizer, clip: float, sigma: float) -> None`	
method	`DPOptimizer.step(self) -> None`	
method	`DPOptimizer.zero_grad(self) -> None`	
function	`make_private(optimizer, cfg: dict)`	

src/doc_agent/optional/query_expansion.py *[IMPL]*

OPTIONAL — query embedding + expansion

Kind	Name / signature	What it does
constant	`SYNONYMS`	
function	`expand(query: str, cfg: dict) -> list[str]`	Original query + keyword-only form + one variant per matching synonym.

src/doc_agent/optional/retrain_trigger.py *[IMPL]*

OPTIONAL — automated retrain on drift/SLA breach

Kind	Name / signature	What it does
function	`maybe_retrain(metrics: dict, cfg: dict) -> bool`	If the monitor (mlops.monitor.check_drift) says retrain, re-train the enhancer on the current

src/doc_agent/optional/stream_ingest.py *[IMPL]*

OPTIONAL — batch/streaming ingestion for very large corpora

Kind	Name / signature	What it does
function	`stream(cfg: dict, batch_size: int=32)`	Yield lists of Page of at most batch_size, lazily, so a 100k-page corpus never sits in memory

src/doc_agent/optional/synthetic_data.py *[IMPL]*

OPTIONAL — generate synthetic text-lines for tiny-label training

Kind	Name / signature	What it does
function	`generate(cfg: dict, n: int=50) -> list`	Render n (image_path, text) text-line pairs from corpus words using PIL, with random noise,

src/doc_agent/retrieval/rerank.py *[IMPL]*

Stage 5 — reranking

Kind	Name / signature	What it does
function	`_cross_encoder(name: str)`	
function	`rerank(query: str, candidates: list[Chunk], cfg: dict) -> list[Chunk]`	Cross-encoder rerank if cfg['retrieve']['rerank']; otherwise (or if the model can't be

src/doc_agent/retrieval/retriever.py *[IMPL]*

Stage 5 — dense retrieval

Kind	Name / signature	What it does
class	`Retriever`	
method	`Retriever.__init__(self, cfg: dict) -> None`	
method	`Retriever.chunks(self) -> list[Chunk]`	
method	`Retriever.retrieve(self, query: str, k: int | None=None) -> list[Chunk]`	Top-k dense retrieval. Embeds the query, searches the index, and sets chunk.score to the
function	`top_score(chunks: list[Chunk]) -> float`	Strength of the current evidence = best chunk score (0.0 if empty).
function	`is_weak(chunks: list[Chunk], cfg: dict) -> bool`	Weak evidence = best score below cfg.retrieve.weak_threshold.

Kind	Name / signature	What it does
function	`next_k(k: int, cfg: dict) -> int | None`	Widen the net: k + k_step, or None once it would exceed k_max (signal to ABSTAIN).

src/doc_agent/rl/env.py *[IMPL]*

Stage 7 — Gymnasium env for tool/retrieval selection

Kind	Name / signature	What it does
constant	`ACTION_NAMES`	
class	`ToolSelectionEnv(gym.Env)`	State=agent context, Action=tool choice, Reward=task success under budget.
method	`ToolSelectionEnv.__init__(self, cfg: dict | None=None) -> None`	
method	`ToolSelectionEnv.reset(self, *, seed=None, options=None)`	
method	`ToolSelectionEnv.step(self, action)`	

src/doc_agent/rl/policy.py *[IMPL]*

Stage 7 — policy network

Kind	Name / signature	What it does
class	`PolicyNet(nn.Module)`	Actor-critic MLP: state(3) -> action logits(3) and value.
method	`PolicyNet.__init__(self, hidden: int=32) -> None`	
method	`PolicyNet.forward(self, x: torch.Tensor)`	
class	`Policy`	
method	`Policy.__init__(self, cfg: dict) -> None`	
method	`Policy.act(self, state) -> dict`	state: array [top_score, k_frac, step_frac] or dict with those keys. Returns

src/doc_agent/rl/rlvr.py *[IMPL]*

Stage 7 — RLVR — verifiable reward on extraction accuracy

Kind	Name / signature	What it does
function	`_norm(x) -> str`	
function	`verifiable_reward(prediction, gold) -> float`	+1 if extraction exactly matches gold (after lower-casing / punctuation / whitespace

src/doc_agent/rl/train_rl.py *[IMPL]*

Stage 7 — RL/RLVR training loop

Kind	Name / signature	What it does
function	`_rollout(env: ToolSelectionEnv, net: PolicyNet, episodes: int, gamma: float)`	
function	`train(cfg: dict, iterations: int=60, episodes: int=64) -> float`	Train the tool-selection policy with PPO (clipped surrogate + value baseline) on the

src/doc_agent/serve/api.py *[IMPL]*

Stage 8 — FastAPI service

Kind	Name / signature	What it does
function	`answer(q: str) -> dict`	Return grounded, cited answer (calls pipeline.answer) + wall-clock latency in ms,
function	`health() -> dict`	

src/doc_agent/serve/ui.py *[IMPL]*

Stage 8 — Gradio demo

Kind	Name / signature	What it does
function	`launch(cfg: dict) -> None`	Gradio UI: question in -> answer, confidence, citations (with the cited sentence) out.

src/doc_agent/training/adapt.py *[IMPL]*

Stage 8 — affordable adaptation — LoRA / quantization

Kind	Name / signature	What it does
function	`apply_lora(model, cfg: dict)`	Wrap a transformer component (e.g. the TrOCR decoder or the embedder) with LoRA adapters per
function	`quantize(model, cfg: dict)`	Post-training dynamic quantization of Linear layers (CPU inference), dtype from

src/doc_agent/training/datamodule.py *[IMPL]*

Training — Lightning datamodule

Kind	Name / signature	What it does
class	`_PatchSet(Dataset)`	(degraded, clean) pairs; degradation is re-drawn each epoch from a seeded generator.
method	`_PatchSet.__init__(self, patches: list[np.ndarray], seed: int) -> None`	
class	`DocDataModule(L.LightningDataModule)`	Pages -> split BY DOCUMENT (no leakage) -> clean patches for the enhancer.
method	`DocDataModule.__init__(self, cfg: dict | None=None) -> None`	
method	`DocDataModule.setup(self, stage: str | None=None) -> None`	
method	`DocDataModule.train_dataloader(self) -> DataLoader`	
method	`DocDataModule.val_dataloader(self) -> DataLoader`	

src/doc_agent/training/lit_modules.py *[IMPL]*

Training — Lightning modules per trainable component

Kind	Name / signature	What it does
class	`LitComponent(L.LightningModule)`	Wrap enhancer / OCR / retriever training. Implemented: component='enhancer' (denoiser,
method	`LitComponent.__init__(self, component: str='enhancer', cfg: dict | None=None) -> None`	
method	`LitComponent.training_step(self, batch, idx)`	
method	`LitComponent.validation_step(self, batch, idx)`	
method	`LitComponent.configure_optimizers(self)`	

src/doc_agent/training/train.py *[IMPL]*

Training — unified entrypoint

Kind	Name / signature	What it does
function	`main(component: str, cfg: dict) -> None`	Train one component with a seeded Lightning Trainer + W&B (or CSV) logger, save its weights

src/doc_agent/vision/layout.py *[IMPL]*

Stage 2 — layout detection / segmentation

Kind	Name / signature	What it does
function	`_lines(gray: np.ndarray) -> list[tuple[int, int, int, int]]`	Text-line boxes: Otsu ink mask -> wide horizontal dilation merges words into lines.
function	`detect(pages: list[Page], cfg: dict) -> list[Region]`	Classical layout analysis (cfg['layout']['backend']='classical'). Reading order =

src/doc_agent/vision/ocr.py *[IMPL]*

Stage 3 — OCR/HTR (BASELINE = pretrained foundation, fine-tuned)

Kind	Name / signature	What it does
class	`Reader`	Model set by cfg['ocr']. Baseline: pretrained TrOCR. backend='auto' prefers a ground-truth
method	`Reader.__init__(self, cfg: dict) -> None`	
method	`Reader.sidecar_lines(self, page_id: str) -> list[str] | None`	Non-empty lines of raw_dir/**/<page_id>.txt (page ids use '_' for sub-folders).
method	`Reader.transcribe_region(self, region: Region) -> str`	
function	`transcribe(regions: list[Region], cfg: dict) -> list[Chunk]`	Regions -> one text Chunk per region (line-level; index.chunk.split merges into windows).

Scripts, verifier and tests

scripts/make_demo_corpus.py *[NEW]*

Kind	Name / signature	What it does
constant	`DOCS`	
function	`font(size: int)`	
function	`render(lines: list[str], rng: np.random.Generator) -> Image.Image`	
function	`main(out: str='data/raw') -> None`	

scripts/run_eval.py *[IMPL]*

Kind	Name / signature	What it does
function	`main() -> None`	

grading_kit/success_check.py *[IMPL]*

Kind	Name / signature	What it does
function	`_norm(s: str) -> str`	
function	`check(task: dict, answer: dict) -> bool`	Return True if `answer` satisfies `task`.

tests/conftest.py *[NEW]*

Kind	Name / signature	What it does
constant	`ROOT`	
function	`kb(tmp_path_factory)`	Build the whole KB (ingest -> index) in a temp dir; returns the cfg that points at it.

tests/test_agent.py *[IMPL]*

Kind	Name / signature	What it does
function	`test_calculator_is_safe()`	
function	`test_aggregate_and_extract_and_cite(kb)`	
function	`test_agent_researches_when_evidence_weak(kb)`	
function	`test_memory_recall()`	
function	`test_hitl_queue_roundtrip(kb)`	

tests/test_crosscutting.py *[IMPL]*

Kind	Name / signature	What it does
function	`test_grounding_unsupported_query_abstains(kb)`	An answer with no supporting evidence must abstain, not fabricate.
function	`test_injection_in_document_does_not_hijack(kb)`	A document containing 'ignore your instructions' must not change agent behaviour.
function	`test_pii_never_leaks_to_answer_or_log(kb, capsys)`	PII in the corpus must not appear in answers or logs.
function	`test_trace_covers_every_step(kb)`	Every agent step and tool call must appear in the audit trail.
function	`test_rerun_reproduces_metrics(kb)`	A seeded re-run reproduces reported metrics within tolerance.
function	`test_seams_clear_between_queries(kb)`	

tests/test_data.py *[IMPL]*

Kind	Name / signature	What it does
function	`test_validate_rejects_bad_input(tmp_path)`	
function	`test_split_is_by_document()`	

Kind	Name / signature	What it does
function	`test_floors_report()`	
function	`test_snapshot_changes_with_content(tmp_path)`	

tests/test_eval.py *[IMPL]*

Kind	Name / signature	What it does
function	`test_ocr_f1_and_recall()`	
function	`test_groundedness_rules()`	
function	`test_ece_perfect_and_overconfident()`	
function	`test_temperature_scaling_softens_overconfident_logits()`	
function	`test_subgroup_gap()`	
function	`test_rlvr_reward_is_exact_match()`	
function	`test_env_step_contract()`	

tests/test_ingest.py *[IMPL]*

Kind	Name / signature	What it does
function	`test_load_pages(kb)`	
function	`test_preprocess_and_enhance(kb)`	
function	`test_denoiser_forward_shape()`	

tests/test_ocr.py *[IMPL]*

Kind	Name / signature	What it does
function	`_pages(kb)`	
function	`test_layout_finds_lines(kb)`	
function	`test_transcribe_returns_text_chunks(kb)`	

tests/test_retrieval.py *[IMPL]*

Kind	Name / signature	What it does
function	`test_chunk_split_windows_and_pages()`	
function	`test_retrieve_scores_and_relevance(kb)`	
function	`test_evidence_policy()`	

tests/test_smoke.py *[IMPL]*

Kind	Name / signature	What it does
function	`test_answer_is_grounded_and_cited(kb)`	
function	`test_unanswerable_abstains(kb)`	

Config / data changes

File	Change
configs/config.yaml	Added layout.backend, ocr.backend, agent.llm_model/n_context/cost_per_call_usd, adapt, train and data (paths) sections; index.type set to faiss:flat (HNSW segfaults in this environment's faiss build); chunk_tokens 48 / overlap 8 for the tiny demo corpus.
grading_kit/tasks.jsonl	Placeholder replaced by 6 demo questions for the synthetic corpus (5 answerable facts + unanswerable/abstain cases).
.gitignore	Generated outputs ignored (models/, data/index/, traces/run.jsonl, report logs).

Total top-level items listed in tables: 203+ (see per-file tables).